Arunabha Majumder



Mechanical & Robotics R&D Engineer — Actuator Development, Controls & Hardware Systems

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Professional Profile

Mechatronics and Mechanical R&D Engineer with practical experience in product development, designing, prototyping, and testing dynamic electromechanical systems, variable-stiffness actuators, and robotic mechanisms. Proven ability to bridge CAD modeling, analytical kinematic/stiffness analysis, dynamic control implementation, and sensor integration into functional hardware solutions.

Core Engineering Competencies

Design & Modeling: Mechanical Product Development · Actuator & Robotic Mechanisms · CAD · FEA
Robotics & Controls: Compliant Actuation · Analytical Modeling · Control Systems · Automation
Testing & Hardware: Prototype Fabrication · Sensor Integration · Custom Test-Rig Development · Data Analysis · Biomechanical Analysis

Professional Experience

Research Engineer — Robotics & Actuator Development (PhD Project: VIEXO) *2023 – Present*
Aalborg University *Aalborg, Denmark*

- **Cut shoulder exoskeleton energy consumption by >25%** by designing a hybrid actuation architecture that coordinates a parallel spring with a 6 Nm active motor, offloading quasi-static gravity torque across the assistance cycle and won **Best Paper Award, IFToMM ISRM 2026**.
- **Reduced user anterior deltoid muscle effort by >15% MVC** by developing and deploying an Assist-as-Needed (AAN) hybrid controller on functional prototype hardware, validated on human subjects.
- **Eliminated the need for an external force/torque sensor** in a 5-bar parallel robot by deriving a sensorless force-estimation method from the actuator torque–deflection characteristic, reducing system cost, mass, and integration complexity while enabling real-time human-arm impedance measurement.
- **Validated analytical stiffness models with an exceptionally low root-mean-square error**(<10% error) against experimental hardware by mapping variable-stiffness actuator (VSA) states into Cartesian workspace stiffness, de-risking the mechanism before fabrication.
- **Designed and executed a 12-participant human trial** across 3 assistance modes and 2 load conditions, covering statistical power justification, EMG and motion-capture instrumentation, ethics protocol, and end-to-end data pipeline, producing the validation dataset for two journal submissions.
- **Built the full prototype stack from concept to tested hardware** — SolidWorks CAD and FEA, mechanical fabrication, motor/encoder/force-sensor and DAQ integration, embedded control, and custom automated test rigs.

M.Tech Researcher — Actuator Systems (Master's Thesis) *2020 – 2022*
CSIR-Central Mechanical Engineering Research Institute (CMERI — National Laboratory, Govt. of India) Durgapur, India

- **Reduced dynamic position-tracking error to 0.3% – 0.78%** by designing and implementing a Neural Network-based gain-scheduled PID controller for a highly non-linear pneumatic actuator, outperforming the classical PID baseline under varying axial load.
- **Fabricated pneumatic artificial muscle actuators in-house from raw materials** — valves, pressure sensors, and DAQ — to characterise load–displacement behaviour and derive an empirical actuator model.

B.Tech Project (Mechanical Engineer) — Rehabilitation Robotics (Bachelor's Capstone) *2018 – 2019*
Siddaganga Institute of Technology *Tumakuru, India*

- **Delivered an award-winning upper-limb rehabilitation exoskeleton** that secured the Best Major Project Award and a 100,000 INR TATA Technologies Innovation Grant, leading CAD design, fabrication, and user testing end to end.
  Youtube Link: (https://www.youtube.com/watch?v=q5Ystz9NCpQ&t=7s)

Education

Ph.D. Candidate in Mechatronics & Robotics *2023 – Present*
Aalborg University, Department of Materials and Production — Aalborg, Denmark

M.Tech in Mechatronics *2020 – 2022*
AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI) — Durgapur, India

B.Tech in Mechanical Engineering *2015 – 2019*
Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU) — Tumakuru, India

Engineering Tools & Softwares

- **Software & Programming:** MATLAB/Simulink, Python, LabVIEW (basics), Arduino, LaTeX, Adobe Illustrator, Adobe Photoshop, Procreate (basics).
- **CAD & Modeling:** SolidWorks, Fusion 360, Finite Element Analysis (FEA), Design for Assembly (DFA), 3D Printing (Bambu Lab, Ultimaker Cura).
- **Instrumentation & Testing:** Qualisys Motion Capture, Forsentec Force/Torque Sensors, Pneumatic Control, Custom Rig Fabrication.

Selected Publications

- **Majumder, A.**, Wagner, J.W., Zhu, Y., Oliveira, A.S., and Bai, S. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." *Robotics and Mechatronics: ISRM 2026*, MMS, vol. 213, Springer. (**Best Research Paper Award Winner**)
- **Majumder, A.**, Sarkar, D., Chakraborty, S., Singh, A., Roy, S.S., and Arora, A. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." *IEEE CONECCT*, 2022.

Awards

- **Best Research Paper Award, 9th IFToMM International Symposium on Robotics and Mechatronics (2026)** — awarded for hybrid-actuated exoskeleton design.
- **Certified SolidWorks Associate (CSWA)** — Mechanical Design.
- **Best Major Project Award** — Department of Mechanical Engineering (2019).
- **TATA Technologies Innovation Award** — INR 100,000 Grant (2019).

Languages Known

1. English (CEFR: C1), 2. Bengali (Native), 3. Hindi (oral proficiency)

Hobbies and Extracurricular

- **Engineering Design & Prototyping:** Led end-to-end development of a custom-fabricated Go-Kart from the ground up (2017) and collaborated within a multidisciplinary team to engineer an eco-friendly human-electric hybrid vehicle for the Efficycle competition (2018).
- **Visual Arts & Illustration:** Professionally trained visual artist holding a Diploma in Fine Arts (Painting) from Bangiya Sangeet Parishad; leverage advanced sketching, painting, and graphic design capabilities to create high-impact illustrations and technical visuals.

References

- **Shaoping Bai, PhD,** Professor
  Department of Materials and Production, Aalborg University, Denmark. Email: sbh@mp.aau.dk
- **Lars Rosgaard Jensen,** Associate Professor
  Head of Section, Physics, Mechanics and Materials; Head of Labs, Department of Materials and Production, Aalborg University, Denmark
- **Anderson de Souza Castelo Oliveira,** Associate Professor
  Department of Materials and Production, Aalborg University, Denmark